

LUKOYANOV, Russian Federation

WMO: 276650

Lat: 55.033N

Lon: 44.500E

Elev: 219

StdP: 98.72

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -26.0	(c) -22.9	(d) -28.8	(e) 0.3	(f) -25.4	(g) -25.6	(h) 0.4	(i) -22.5	(j) 7.5	(k) -5.2	(l) 6.8	(m) -6.3	(n) 1.6	(o) 350	(p) 0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.2	(c) 30.0	(d) 20.0	(e) 28.0	(f) 19.1	(g) 26.3	(h) 18.3	(i) 21.1	(j) 27.5	(k) 20.3	(l) 26.3	(m) 19.3	(n) 24.8	(o) 2.1	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.0	14.2	23.8	18.1	13.4	23.1	17.2	12.6	22.2	62.1	27.7	59.1	26.6	56.0	24.8	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.4	5.8	5.0	-28.2	32.3	4.0	2.6	-31.2	34.1	-33.5	35.6	-35.8	37.1	-38.7	38.9
			-28.4	22.5	4.0	1.1	-31.3	23.3	-33.6	23.9	-35.8	24.6	-38.7	25.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.0	-9.6	-9.0	-3.4	5.9	13.3	16.9	19.5	17.6	11.8	4.8	-2.2	-7.3	(6)
(7)		DBStd	11.68	6.83	6.78	4.69	5.30	5.10	4.29	3.72	4.21	4.37	4.53	5.38	6.57	(7)
(8)		HDD10.0	2826	608	531	414	144	25	3	0	1	29	168	365	537	(8)
(9)		HDD18.3	5043	867	764	673	372	168	76	31	64	199	418	614	795	(9)
(10)		CDD10.0	987	0	0	0	23	129	211	294	237	84	9	0	0	(10)
(11)		CDD18.3	161	0	0	0	0	13	34	67	42	4	0	0	0	(11)
(12)		CDH23.3	1302	0	0	0	1	121	283	524	344	29	0	0	0	(12)
(13)		CDH26.7	349	0	0	0	0	20	71	146	111	3	0	0	0	(13)
(14)	Wind	WSAvg	2.2	2.5	2.5	2.4	2.4	2.2	2.0	1.8	1.8	1.9	2.3	2.5	2.5	(14)
(15)	Precipitation	PrecAvg	578	39	30	32	34	39	67	71	65	59	51	45	46	(15)
(16)		PrecMax	803	85	57	69	76	114	165	163	155	152	109	92	115	(16)
(17)		PrecMin	343	11	9	1	7	18	15	8	5	9	15	9	11	(17)
(18)		PrecStd	94	15	14	17	21	19	37	30	35	34	26	20	25	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.0	2.5	9.8	22.6	29.0	31.6	33.6	33.7	26.6	18.8	10.6	5.1	(19)
(20)			MCWB	1.0	1.1	5.8	12.4	17.7	19.1	19.9	20.3	16.4	12.5	8.8	4.4	(20)
(21)		2%	DB	0.9	1.5	6.1	19.7	26.5	28.6	29.9	29.7	23.6	15.4	7.6	2.8	(21)
(22)			MCWB	0.2	0.5	3.0	11.7	16.6	19.4	20.4	19.8	15.6	11.2	6.6	2.0	(22)
(23)		5%	DB	0.2	0.7	4.0	17.1	24.2	26.6	28.0	26.8	21.2	13.0	5.8	1.3	(23)
(24)			MCWB	-0.4	-0.1	1.8	10.3	15.6	18.5	19.7	18.7	14.9	10.0	4.9	0.6	(24)
(25)		10%	DB	-0.7	0.0	2.5	14.5	21.9	24.6	26.3	24.6	18.8	11.1	4.1	0.3	(25)
(26)			MCWB	-1.3	-0.6	0.9	9.0	14.5	17.7	19.0	17.8	13.8	9.0	3.4	-0.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.4	1.6	6.4	13.9	19.4	21.7	22.6	21.6	18.1	13.5	8.9	4.6	(27)
(28)			MCDWB	1.7	2.1	9.6	20.1	26.9	28.5	28.7	28.1	23.9	16.2	10.1	5.1	(28)
(29)		2%	WB	0.3	0.7	3.4	12.1	17.8	20.6	21.3	20.6	16.7	11.8	6.8	2.2	(29)
(30)			MCDWB	0.6	1.2	5.4	18.1	24.5	26.6	27.3	27.5	21.8	14.9	7.6	2.7	(30)
(31)		5%	WB	-0.3	0.1	2.1	10.8	16.5	19.4	20.5	19.6	15.3	10.4	5.1	0.8	(31)
(32)			MCDWB	0.1	0.6	3.5	16.3	22.7	24.7	26.4	25.8	19.7	12.6	5.7	1.2	(32)
(33)		10%	WB	-1.3	-0.6	1.1	9.4	15.1	18.3	19.6	18.6	14.2	9.2	3.4	-0.1	(33)
(34)			MCDWB	-0.7	0.0	2.3	14.1	20.7	23.3	25.1	23.5	18.1	11.0	4.0	0.3	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	5.9	6.6	8.5	10.0	9.3	9.2	9.2	8.0	5.8	4.2	4.6	(35)
(36)			MCDBR	4.1	4.8	7.7	11.9	12.5	11.7	11.4	12.1	11.6	9.3	5.2	3.5	(36)
(37)			MCWBR	4.1	4.4	5.3	6.3	5.7	5.1	4.6	4.9	5.6	5.7	4.7	3.5	(37)
(38)		5% WB	MCDWB	4.1	4.6	6.8	11.2	11.5	10.4	10.1	10.5	10.3	8.1	4.9	3.2	(38)
(39)			MCWBR	4.2	4.4	4.9	6.3	5.9	5.2	4.6	4.9	5.7	5.7	4.7	3.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.283	0.319	0.350	0.410	0.383	0.377	0.400	0.422	0.371	0.350	0.314	0.285	(40)	
(41)		taud	2.093	2.051	2.041	2.113	2.323	2.367	2.315	2.239	2.367	2.330	2.222	2.080	(41)	
(42)		Ebn at Noon	619	728	800	806	860	868	838	789	787	705	586	516	(42)	
(43)		Edh at Noon	65	98	126	137	117	114	118	120	92	76	58	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.73	1.66	3.25	4.12	5.50	5.70	5.73	4.56	2.83	1.35	0.54	0.40	(44)	
(45)		RadStd	0.12	0.23	0.38	0.39	0.39	0.53	0.54	0.46	0.40	0.21	0.10	0.11	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.64	N/A	N/A	N/A	N/A	+0.60	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.59	N/A	N/A	N/A	N/A	N/A	-125	-173	N/A	N/A

Nomenclature: See separate page